



Auburn-USDA Final Report

Harvey Mudd College Engineering Clinic 2024-2025

Pierce Gruber '25 (TL-S)
Alisha Chulani '25 (TL-F)
Julia Hansen '25
Dimitri Avila '25
Leilani Elkaslasy '26 (S)
Emma Angel '26 (S)
Nina Jobanputra '26 (F)
Amy Liu '26 (F)

Project Advisors: Professor Victor Shia (S) and Professor Matthew Spencer (F)
Project Liaisons: Dr. Elaine Backus, Dr. Kathryn Reif, Dr. Anastasia Cooper, Andrew Dowell

Abstract

This report details work done by Harvey Mudd College's 2024-25-25 Auburn-USDA Engineering Clinic team to modernize and improve electropenetograph technology. Electropenetographs are used by entomologists to study the feeding behavior of insects, and this clinic project investigates solutions to increase the signal-to-noise ratio, integrate wireless communication, and enhance the user-experience of this device. This project is a continuation of work done by 2023-24 Engineering and CS clinic teams, with improvements being made on the analog, digital, and software portions of the project.

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Section 1: Introduction

Piercing-sucking insects are dangerous pests, and their danger may grow as the earth's climate warms. Crop and cattle damage caused by piercing-sucking insects, such as aphids and ticks, can lead to large economic losses and tragic human suffering. For instance, aphid damage to crops can cost \$104 million to \$1 trillion annually [1]. Furthermore, due to climate change, tick numbers continue to increase [2].

Understanding the feeding behavior of piercing-sucking insects can help reduce the threat they pose. Electropenetographs (EPGs) are a tool used by entomologists to study arthropod feeding behavior. EPGs require a wire to be affixed to an insect and to a substrate, like a plant or animal, which results in the insect completing a circuit when it pierces a plant or animal with its mouthparts. EPGs then measure the electrical waveforms produced by such a circuit, allowing entomologists to visualize feeding waveforms [3].

The 2024-25 Auburn-USDA clinic team worked to improve EPG hardware. This report describes their work. This section introduces some background on EPGs, some prior work completed by clinic teams in 2023-24, and the organization of the report.

1.1 - Electropenetograph Background and Problem Statement

Figure 1 shows how a plant and insect are connected for EPG measurements, and Figure 2 shows a simplified electrical model of the insect-plant system surrounding EPG hardware. A voltage stimulus, V_s , is applied to the bug-plant system, represented by a variable voltage and resistance. V_{emf} represents the flow of ionized fluids through the insect's mouthparts and R_a represents the interface between the insect and the plant. As the bug feeds, the V_{emf} and R_a values change, which in turn alter the voltage across the R_i resistor. Entomologists interpret the R_i voltage to understand piercing-sucking behaviors.

The voltage across R_i is small, typically microvolts, so an amplifier is required to measure the voltage. The amplifier must also carefully reject noise. One representative type of EPG amplifier, called the Legacy AC-DC EPG, is housed in a large (13x9x4 inch) control box. Entomologists must set up the system inside a grounded Faraday cage to limit noise for readable measurements. Figures 3 & 4 show the control box and the Faraday cage setup.

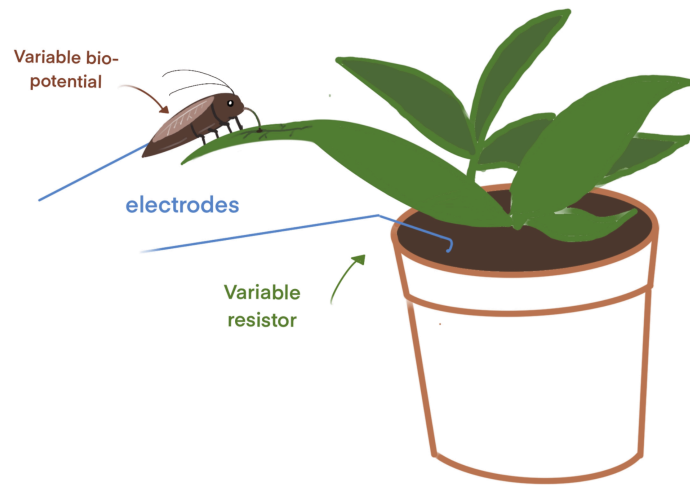


Figure 1: Physical bug-plant system set-up showing where probes of EPGs hook up

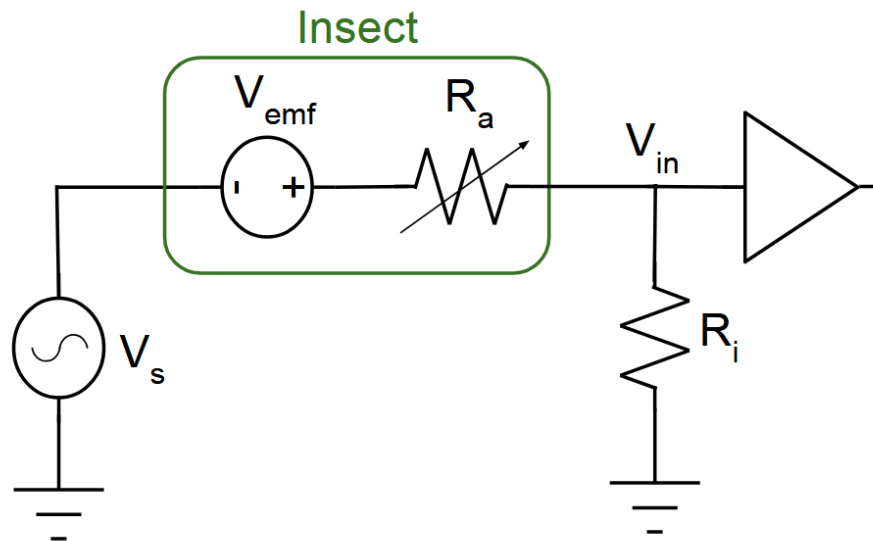


Figure 2: Electrical schematic of EPGs

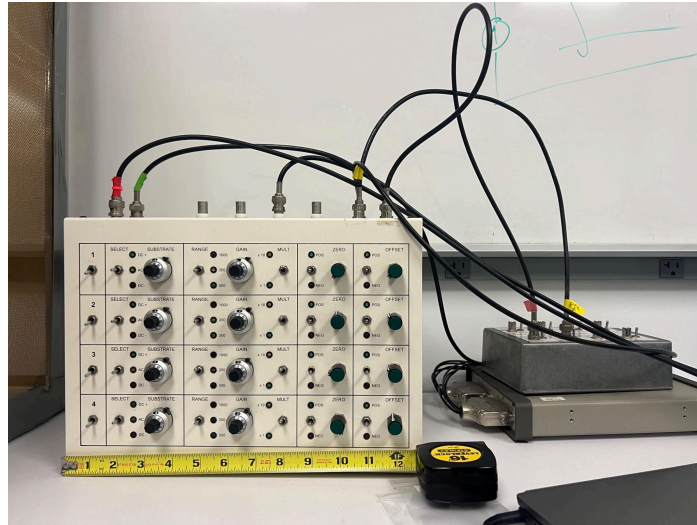


Figure 3: EPG Control Box for adjusting gain and offset settings on EPGs



Figure 4: Faraday cage in which plant and bug need to be set up to allow for grounding and reduced noise.

A laptop can be connected to the EPG control box and as the bug feeds, voltage values and waveforms are displayed live on a software called WinDaq. Entomologists can view and make annotations on these waveforms live, as well as save and export the data for later analysis.

The goal of this clinic project is to redesign EPG amplifiers and enclosures to extend its applications beyond plant-feeding aphids to blood-sucking arthropods that attach on live animals. Taking advantage of modern digital hardware can help make the amplifier smaller, reduce the noise picked up by the amplifier to remove the need for the Faraday cage, and enable the EPG to use battery power and wireless data transfer to make the EPG portable. Additionally, this clinic project will also look to create a seamless software with the improved analog-digital product.

This document describes efforts made by the 2024-25 Auburn-USDA Engineering Clinic team to improve EPG amplifiers. This technical progress continues the work done by both the 2023-24 Engineering and CS clinic teams. Some of the work this semester specifically addresses the problems in Prototype 6. These efforts were motivated by the following problem statement:

For the past 15 years, entomologists have relied on the AC-DC Electropenetrograph (EPG) to study the behavior of plant-feeding aphids in a lab setting. There is an urgent need for a new device to study the behavior of blood-sucking arthropods in a field setting. This new device needs to be portable enough to be comfortably deployed on hosts, but it must also maintain the quality of data that entomologists are familiar with. Researchers at Auburn University and the USDA have tasked this year's clinic team to continue the work of last year's clinic team by refining their digitized EPG to increase the signal-to-noise ratio, integrate wireless communication, and enhance the user-experience. These changes will enable the study of both plant-feeding and blood-sucking arthropods.

1.2: Previous Efforts

This clinic project has been in progress since the fall of 2023-24, during which two clinic teams were commissioned to address similar challenges. The 2023-24 Auburn-USDA Engineering Clinic team, referred to as the 2023-24 Engineering team hereafter, produced prototype EPG amplifiers in pursuit of similar goals as the 2024-25 Auburn-USDA clinic team, referred to as the 2024-25 Clinic team hereafter. The 2023-24 Auburn-USDA Computer Science (CS) clinic team, referred to as the 2023-24 CS team hereafter, worked in tandem with the 2023-24 engineering clinic team to build software to control the 2023-24 engineering team prototypes via a graphical user interface. These prototypes and software were used as a starting point to

inspire the bug fixes and add feature improvements targeted by the 2024-25 Auburn-USDA Engineering clinic team [2].

All of the clinic teams were attempting to produce an improved EPG, which is referred to as a digital EPG. The 2023-24 version of this improved product will be called the 2023-24 digital EPG. Every digital EPG design thus far is composed of an analog board, a digital board, firmware, and frontend.

The 2023-24 Auburn-USDA Engineering Clinic team referred to their prototypes by number – Prototype 1, Prototype 2, etc. – and this report will use those names. Digital EPG prototypes produced by the 2024-25 Auburn-USDA Engineering clinic team will be referred to as the Dual Rail 1, 1a (DR1/DR1a) and Dual Rail 2 (DR2) prototypes and Single Rail 1 prototype (SR1). There are no such numerical versions of the software created by the 2023-24 Auburn-USDA Computer Science Clinic team, so the final software produced by the 2023-24 Auburn-USDA Computer Science Clinic team (which was delivered in May 2024-25) will be referred to as the 2023-24 software or Clinic-v1 Firmware and Frontend, while the software produced by the 2024-25 Auburn-USDA Engineering team will be referred to as the 2024-25 software or Clinic-v2 Firmware and Frontend.

Certain prototypes from the 2023-24 Engineering and CS team had success. Prototype 4 was used to collect data from an insect, and that data matched results from the existing EPG. Prototype 6 was the final prototype, which was built to address issues from Prototype 4, but which ultimately exhibited problems at the close of the 2023-24 clinic year [10]. Some of this year's work looks to solve the problems that Prototype 6 displayed, and Section 3 specifically addresses analog design issues from Prototype 6 and this year's solutions.

1.3: Document Organization

Section 2 will first provide an overview of the system architecture and overarching design decisions that were made that affect all components of the system, such as a power system analysis and EPG testing with a validator board. Section 3 will address analog updates and design improvements made from last year's Prototypes 4 and 6. Section 4 will discuss work on the digital side, starting with digital issues from last year's prototype. This section will address new dev board options, analog board interfacing, bluetooth transmission, and data collection. Section 5 will discuss firmware and software, starting with the remaining bugs from the 2023-24 clinic. This section will also discuss UI/UX improvements and integration with the digital board as discussed in Section 4. Section 6 describes the mechanical setup chosen for this year's prototypes. Section 7 describes the outcomes of site visits 1 and 2 this Spring, including the data gathered on aphids and sharpshooters on both the dual rail and single rail prototypes.

Section 8 provides an overview of the successes of this year’s clinic team in the Fall and Spring, and the future work to be done in the summer, and Section 9 concludes the work.

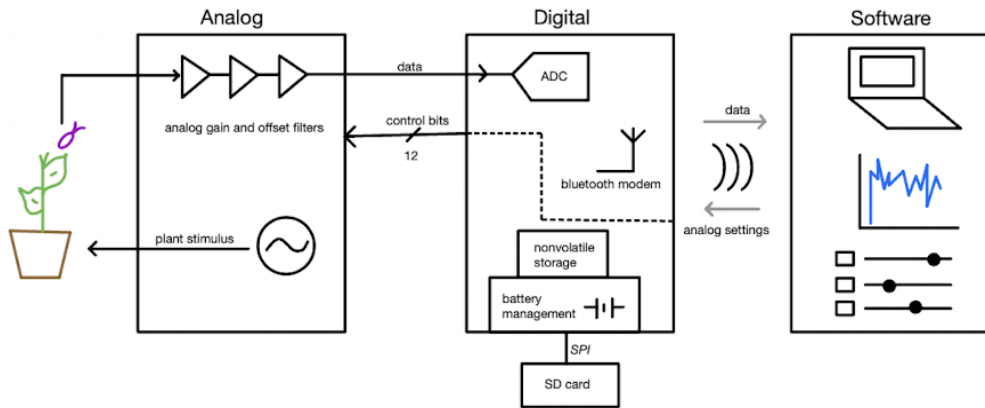


Figure 5: Overall system architecture diagram

Section 2: System Architecture

This section will describe components that span both the analog and digital boards. It includes a power system analysis, a description of the board selection decisions for both analog and digital, wireless communication protocol, and preliminary EPG testing and the validator board.

2.1 - Splitting up the System

Both the 2023-24 and 2024-25 engineering teams split up the system into an analog board, a digital board, and the software. Figure 5 shows an overview of the 2024-25 digital EPG, which highlights this system architecture and the connections between the analog, digital, and software components. It is easy to buy an off-the-shelf digital board, which is why the analog and digital systems are separate boards.

Despite similar architectures, the 2023-24 and 2024-25 engineering teams made different decisions for each component. Both teams designed different analog boards to implement the same gain and filtering process done by Legacy AC-DC EPGs: the Prototype 4 and Prototype 6 analog boards for the 2023-24 engineering team, and the DR1, DR1a, DR2 and SR1 analog boards for the 2024-25 engineering team. The digital boards differed too: the 2023-24 engineering team used a Teensy 4.1 as their digital board, while the 2024-25 engineering team decided to choose the nRF5340 development kit (nRF5340-DK) as the digital board (see Section 4.2).

2.2 - Power System Analysis

The power available to an EPG determines a great deal about its design space. Consequently, design on the 2024-25 digital EPG started with a thorough power system analysis. This analysis consisted of a review of past designs and possible power sources, a study of the conducted noise benefits of using battery power, and a capacity analysis of some common batteries. The analysis was followed by a final power system decision: the DR boards would be powered by two 1S (± 3.7 V) Lithium Polymer (LiPo) batteries and the SR board would be powered by a single 1S (± 3.7 V) LiPo battery.

An important first step in designing the 2024-25 digital EPG system architecture was to review the power system of past boards. One important classification is whether these power systems provide dual rails (positive and negative voltage supplies) or a single rail (positive voltage supply only) for the analog board. The Legacy AC-DC EPG uses a dual rail architecture, and 2023-24's successful Prototype 4 emulated the Legacy AC-DC EPG by using the same dual rail design. The voltage values of the voltage rails differ in these systems, however. The Legacy AC-DC EPGs use ± 12 V rails from wall power, while 2023-24's Prototypes use ± 7.4 V corresponding to two 2S LiPo batteries.

There were many possible ways to power the 2024-25 digital EPG. Since the 2024-2025 digital EPG uses an analog and digital board, both of their power supplies need to be considered. The options for powering the digital board, the nrf5340-DK, are outlined in the User Guide: USB power, a 2.5-5.0 V LiPo battery connector, a 3.0-5.0 V external voltage source, or a 3.0V LiMnO₂ coin cell [6]. The analog board could have been designed to be powered by wall power, which could support either one or two rails, or a variety of LiPo batteries: a single 1S battery (± 3.7 V), two 1S batteries (± 3.7 V), or two 2S batteries (± 7.4 V). It also would have been possible to have the digital board power the analog board or vice versa.

In considering battery power, The 2023-24 final report suggested that a battery-powered system would eliminate some of the conducted interference (the 2023-24 report called this noise) derived from wall-power. To test this, the 2023-24 analog board was hooked up to battery power and then wall power, keeping everything else constant. Figure 6 and 7 show the difference in noise levels using the Fast Fourier Transform (FFT) function on an oscilloscope. The battery powered system decreased overall noise in the signal compared to wall power.

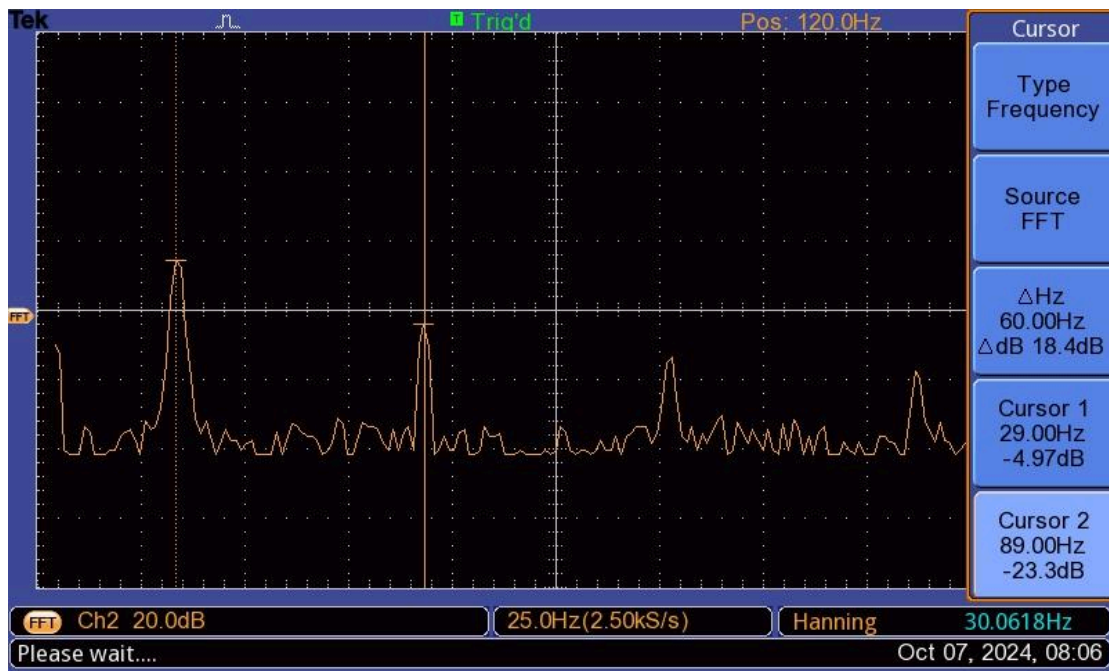


Figure 6. FFT of the output of 2023-24's analog board when powered with wall power.

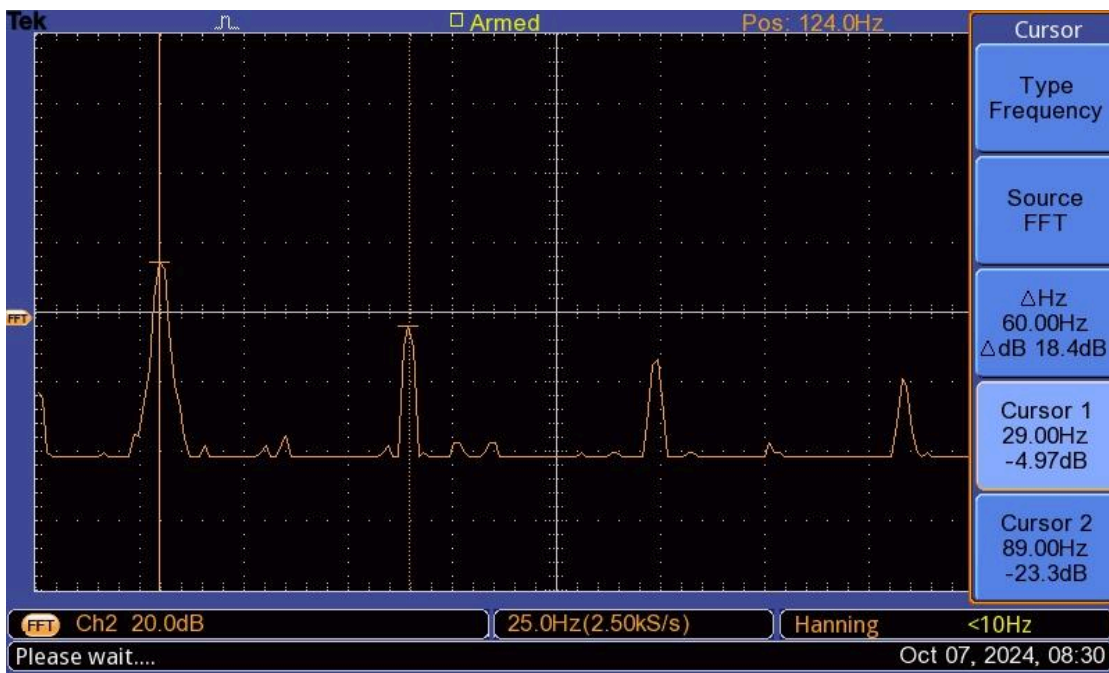


Figure 7. FFT of the output of 2023-24's analog board when powered with battery power.

Additionally, a current analysis was conducted to determine the theoretical operating life of the DR1 DR1a, DR2, and SR1 analog boards. Table 1 shows how many hours each analog board would be available to operate for depending on the size of the battery and estimated total current draw calculated by adding datasheet current estimates for the components used in the design. The two battery options allow entomologists to weigh between two options — longer operation time versus mass — depending on the experimental situation.

This power analysis inspired several decisions. The DR1, 1a, and DR2 boards are dual-rail, like the Legacy AC-DC EPG and Prototype 4, to minimize biological risk. For dual rail boards, the digital and analog boards are powered separately. The analog board is powered through two 1S batteries and supplies power to the digital board as well. Since the individual components in the analog board can all operate within the 3.7 V bounds, two 1S batteries with 8000mAH capacity were used, which are significantly smaller and lighter than the equivalent 2S batteries used by 2023-24 Prototypes.

However, these design decisions highlighted a simpler option: building a single rail board that centers the plant stimulus and amplifier measurement at mid-rail. A prototype, SR1, that uses this single rail architecture is detailed in Section 3.10. SR1 will aspire to have less components, thus opening the possibility of powering the analog board off of the digital board.

	Dual rail 1	Dual rail 1a	Dual rail 2	Single rail 1
Total current consumption	0.03554	0.02466	0.01966	0.01481
450 mAH battery capacity	12.66178953	18.24817518	22.88911495	30.38487508
5200 mAH battery capacity	146.3140124	210.8678021	264.4964395	351.1141121

Table 1: Current analysis for battery life for the different 2024-25 analog board options

2.3 - ADC Analysis

The ADC is used to convert the voltage coming out of the analog board into a form readable by the digital board. It is configured to use a 12-bit resolution and offers two reference options: an internal reference or VDD as the reference. When the internal reference is selected, the ADC core has an input range of ± 0.6 V. In contrast, using VDD as the reference results in an input range of $\pm VDD/4$ on the ADC core. The gain block can be employed to modify the effective input range [9]. For this analysis, the ADC was configured with the internal reference and a gain of $\frac{1}{2}$. With this setup, using a single-ended input (with the negative input grounded), the effective input range was calculated as follows:

$$\text{Input range} = 0.6 \text{ V} / (\frac{1}{5}) = 3.0 \text{ V}$$

It is important to note that the input voltage must remain within the bounds of VSS and VDD, as values outside of this range are not supported. For testing purposes, the ADC's sampling rate was set to 1 sample per second; however the ADC is capable of sampling at rates as high as 200,000 samples per second.

2.4 - Wireless Protocol

One of the necessary specifications for the digital board was to be able to communicate wirelessly to the main computer in order to make the device portable and to reduce conducted noise through the supply. The most simple and power saving way to do this was through implementing Bluetooth. Bluetooth is better than other wireless systems such as Wi-Fi because of the reduced complexity in setting up connections and hardware involved. Specifically Wi-Fi systems require routers which increase the cost of setup and where the device can be placed (outdoors vs indoors). Additionally BLE is best designed for battery operated devices due to its power efficiency. It also provides fast and stable connections through its advertisement and scanning profile (see Section 4.8) [4]. This need for wireless communication was addressed by picking a Nordic board (see Section 4.2 for more details).

2.5 - EPG Testing and the Validator Board

Since insects under test are difficult and time-consuming to prepare, EPGs are instead tested by physically touching the EPG head amp in a procedure called the “touch test.” While being grounded to the Faraday cage, one touches the head amplifier and checks if a positive and negative voltage spike is seen on the output. Figure 10 shows a sample output of a touch test on an Legacy AC-DC EPG. The black signal of this figure is a filtered version of the red signal. The spikes shown in Figure 8 display when the head amp is touched with a human hand while the spikes in Figure 9 show when the head amp is simply grounded to the Faraday cage with jumper cables. These figures show correct functioning of an EPG.

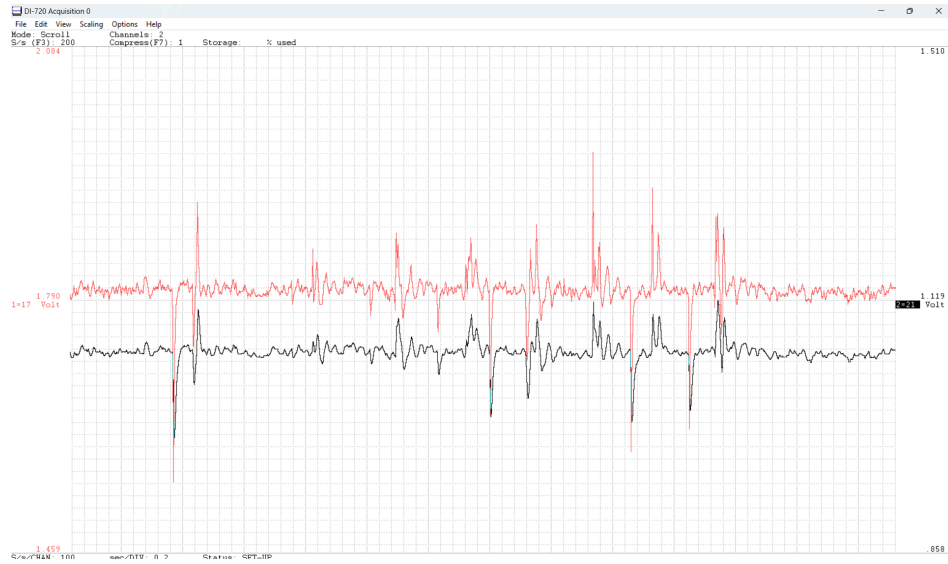


Figure 8: Sample output of Legacy AC-DC EPG finger touch test

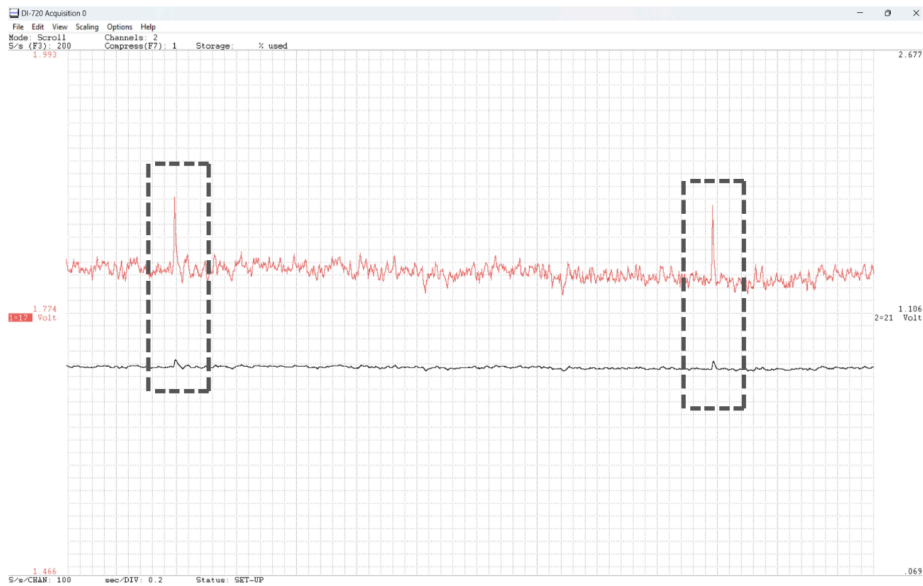


Figure 9: Sample output of Legacy AC-DC EPG touch test, grounding via jumper cables

To test the digital EPG, a standardized version of the touch test was built via a custom PCB called the validator board. This would fit into place where the bug and plant setup is in Figure 2 and 5. Figure 10 shows the final schematic of the validator board. Depending on which switch is toggled, a human touch test, grounding test, or an excitation test (feeding a “bug” waveform into the EPG) is performed.

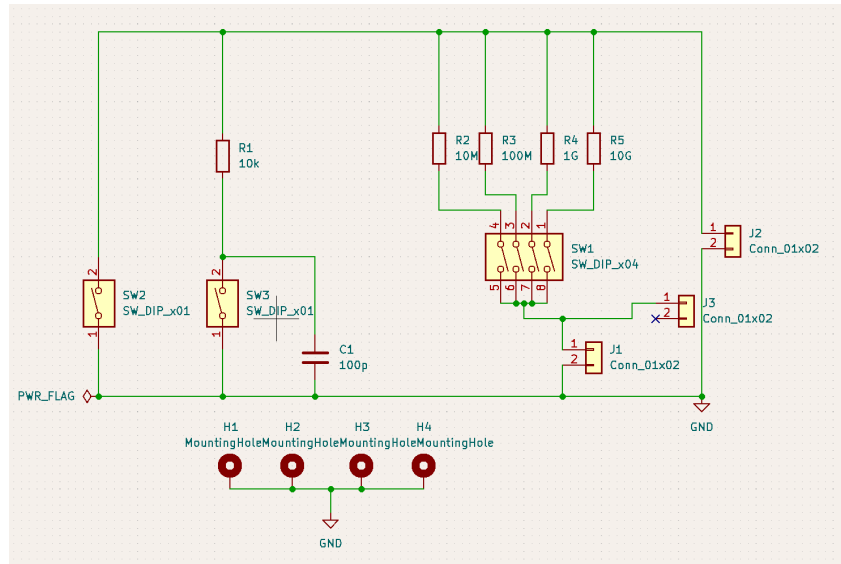


Figure 10: Final schematic of the validator board

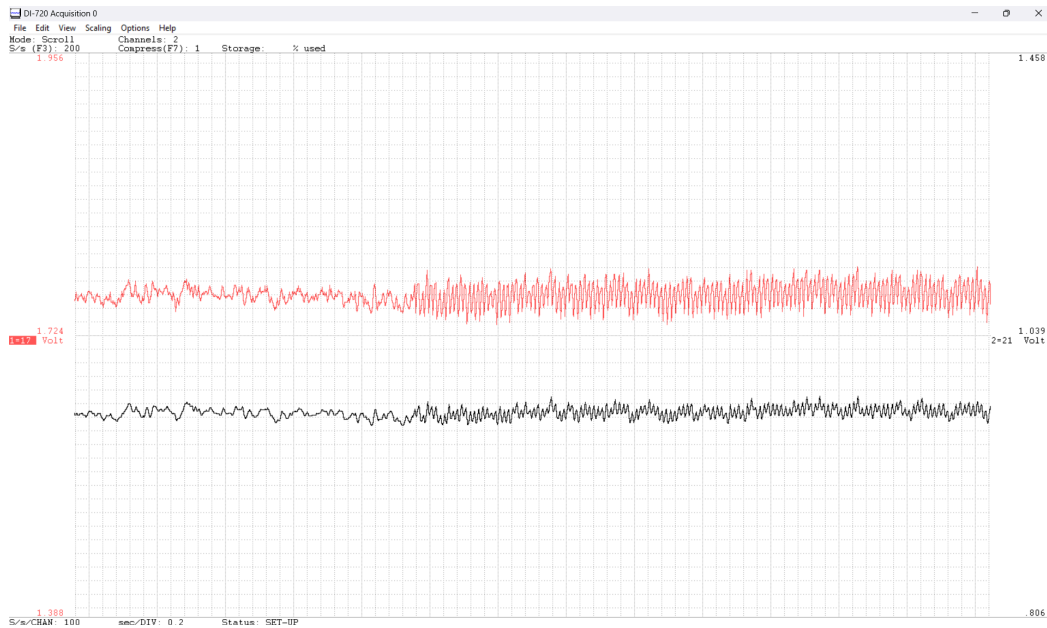


Figure 11: Sample output of excitation tests using validator board with Legacy AC-DC EPG

The validator board was tested with the legacy EPG and produced successful results for traditional excitation tests but not human finger touch tests or grounding tests. Figure 11 shows the outputs of using the validator board's excitation tests on an Legacy AC-DC EPG. This test used a 1Vpp 16 Hz sine wave input, R_a set to 10^7 ohms, and R_i set to 10^6 ohms. It was observed that higher R_a values led to smaller received signals, which accurately matches theory.

However, since this prototype was not able to emulate the human finger touch test or grounding test, it was not used for the purposes of testing the DR boards. Future work would look into revising the board to adequately replicate the touch test as well as emulating the insect's voltage emf.

Section 3: Analog Board

This section will describe the components on the final analog board prototype, DR2, by referencing problems from the 2023-24 team's board and how the 2024-25 team addressed them. Figure 12 shows an abstracted view of the analog board, of which each component will be described in detail.

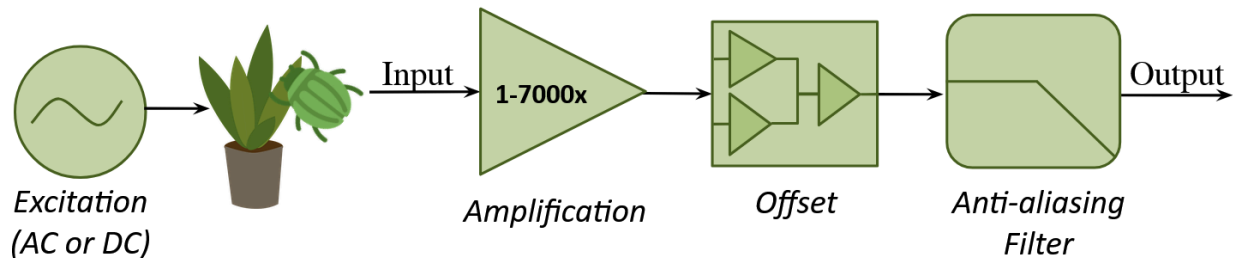


Figure 12: Analog signal chain including excitation voltage

3.1 - Head Stage Amplifier

The head stage amplifier is the first step of the analog amplification chain. Traditionally, this non-inverting amplifier has a fixed gain of 21x. This stage must have a low voltage offset because the offset of this stage is amplified by the whole receive chain (by a gain of up to 7,000x). The stage must also have low input bias current because the input bias current passes through the very large R_i resistance (up to 10 G Ω).

Prototypes 4 and 6 had the OPA4134 as its head stage amplifier configured as a non-inverting amplifier with a fixed gain of 21x. The OPA4134 has an input offset voltage of 3.5 mV and an input bias current of 100 pA [5]. This resulted in large offsets that were cancelled with the offset amplifier stage described in section 3.4. In addition, while the fixed 21x amplification matched the Legacy AC-DC EPG, the 2024-2025 team found that it greatly limited the amplification resolution.

For DR2, this amplifier was replaced by the LMC662CMX, which has a similar input offset voltage but a much lower input offset current, 3 mV and 2 fA respectively [6]. In addition, it was configured as a programmable non-inverting amplifier by interfacing it with a digital

potentiometer, described in section 3.2. This amplifier works in tandem with the digital potentiometer by changing the amplification seen on the signal chain according to the programmed resistor ratio in the potentiometer. With this change, the minimum amplification is no longer 21 but can instead go as low as 1.

3.2 - Digital Potentiometer

Digital potentiometers are electronic components that mimic traditional mechanical potentiometers, allowing programmable control of resistance for signal conditioning. Prototypes 4 and 6 used the MCP4151-103E/SN potentiometer. The MCP4151-103E/SN is a single-channel, single-supply, 256 step (8-bits), 10 K Ω potentiometer, and it was used solely to generate a positive voltage for the offset circuit in the signal chain. The 2024-2025 team found that a new potentiometer chip was needed to serve four new functions: create positive and negative voltages for both the signal chain and excitation offset circuits (Section 3.4), and change resistance ratios for both the head stage (Section 3.1) and excitation amplifiers (Section 3.8).

In DR2, the X9250 was chosen to fulfill these needs. The X9250 is a four channel, dual supply ($\pm 3.3\text{V}$), 256 step (8-bits), 50K Ω digital potentiometer. With its four channels, it can perform its four functions independent of each other. Because it runs on a dual supply, it can generate both positive and negative voltages and be configured in both an inverting or non-inverting amplifier. With its 8 bits of control, it can achieve a theoretical resolution of 0.026V for offset circuits and a resolution as high as 1.004x for amplifiers which is precise enough for the analog board's needs.

3.3 - Programmable Gain Amplifiers

The Programmable Gain Amplifiers (PGAs) from Prototypes 4 and 6 allowed the digital EPG to match the high amplification capacity of the Legacy AC-DC EPG. The first PGA amplifies the signal up to 10x and the second PGA amplifies up to 700x. There were no observed problems with the PGAs during testing, therefore DR2 contains the same PGAs in the same configuration.

3.4 - Offset Circuits

The offset circuit generates a constant voltage and adds it to the signal, effectively “shifting” the signal up or down by that voltage value. In Prototype 4 and 6, one of these offset circuits was used at the end of the signal chain to ensure the insect waveform was in the readable range of the ADC (between 0 and 3.3 V). During testing, this circuit worked as expected and ended up being reused in DR2.

The offset circuit is used twice in DR2: once in the excitation circuit and another in the signal chain. The excitation offset circuit serves to zero-center the AC signal or provide a purely DC plant stimulus, described in Section 3.8. The signal chain offset circuit compensates for any offset gathered from the previous stages and ensures the output insect waveform is within the readable voltage range of the ADC. Although the improved head stage amplifier described in Section 3.1 should reduce the offset throughout the system, DR2 retains this circuit in case unwanted offset is still present in the signal chain.

3.5 - 60 Hz Reject Filter (Removed in 2024-25)

In the 2023-24 analog board, Prototypes 4 and 6 made use of a 7th order low pass filter with a corner frequency of 40 Hz to knock out 60 Hz hum and high frequency noise while retaining bug signals present at a maximum of 30 Hz [Backus and Patterson, 2023-24]. Since most bug waveforms are square waves instead of sine waves [Backus and Patterson, 2023-24], it was found that the filter actually produced significant corner smoothing. This was due to attenuation of higher frequency square wave harmonics, seen in Figure 13. Therefore, the filter was removed in DR2.

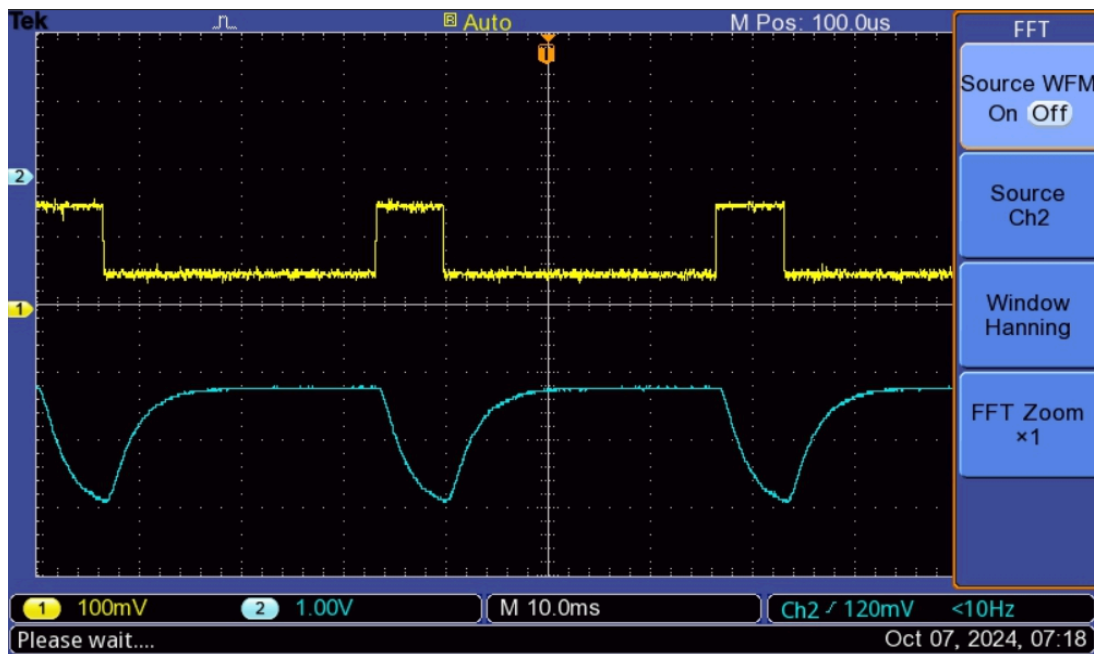


Figure 13. A 30 Hz square wave is fed into Prototype 6 (yellow). The output (blue) shows significant corner smoothing as a result of the harsh 7th order low-pass filter.

3.6 - Anti-Aliasing Filter

The anti-aliasing filter is used to limit frequencies greater than the Nyquist frequency from reaching the ADC, which prevents aliasing from interfering with the insect waveforms. This filter must also avoid the shortcomings of the 60 Hz reject filter discussed in Section 3.5. This means that the anti-aliasing filter must be a low-pass filter, but with a corner frequency that is not within an order of magnitude of the insect square wave frequency. A first-order low-pass filter with a 500 Hz corner frequency was therefore selected to balance preserving insect waveforms while attenuating higher frequencies that might cause aliasing at the ADC.

3.7 - Voltage Regulators

Since the voltage supply may vary based on the type of power source used, voltage regulators are required to stabilize the power before it reaches the board's components that are rated for very specific supply voltages. In Prototypes 4 and 6 there were three regulators: the LD1117S50TR, LT1175CST-5#PBF, and LD1117S33TR that converted the supply voltage down to 5V, -5V, and 3.3V respectively. To power the board, a physical switch had to be engaged to direct the supply voltage to the regulators and, in turn, to the rest of the circuitry. This switch was used to ensure the transient response of the power supply (Figure 14) settled before providing its power to the board. This system was okay if users remembered to set the switch in the off position before turning on the power supply. However, if the switch remained on when the power supply was turned on, the transient response of the power supply would reach the board and immediately fry the negative voltage regulator, shorting the entire board. This caused significant problems during the early testing of Prototype 6.

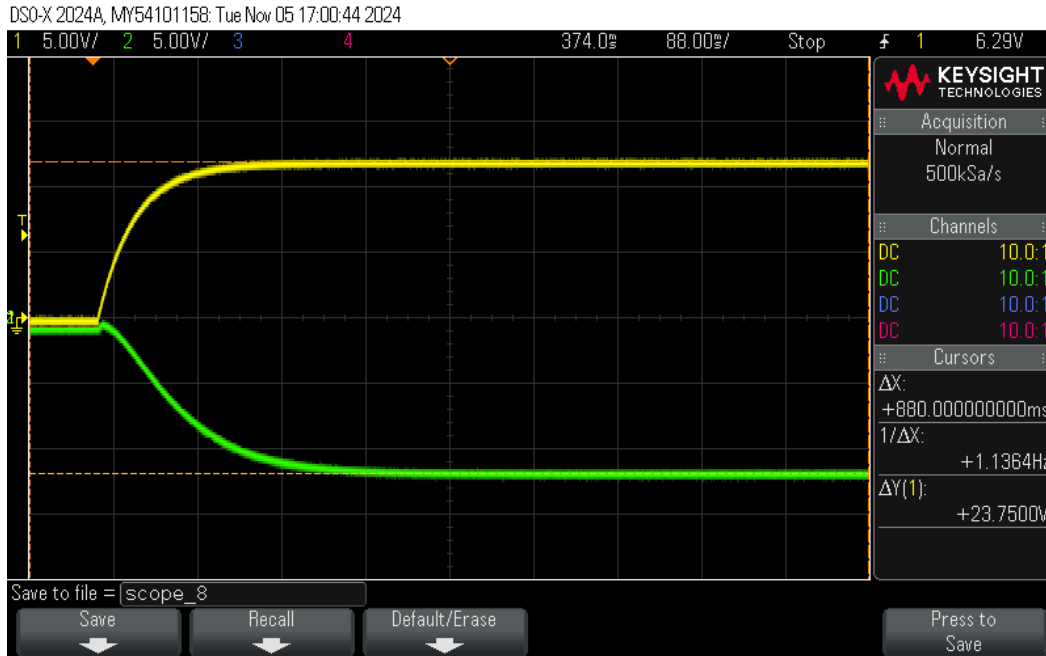


Figure 14: The transient response for the power supply that was used to power the 2023-2024 prototype analog board. The yellow line is the positive rail and the green is the negative rail, both being powered at $\pm 12\text{V}$. The positive rail comes up slightly before the negative rail which was leading the negative regulator to break.

In DR2, two new voltage regulators were picked to prevent the board from being susceptible to this transient power supply behavior: The LT3015EDD-3.3#PBF negative 3.3V regulator and the LP2985-33DBVR positive 3.3V. This negative voltage regulator was chosen due to its fast transient response, current-limiting capabilities, and reverse output protection so as to limit the possibility of supply transients shorting the board. Additionally, both regulators have high input voltage limits up to 16 V, meaning the board will be able to handle large power sources if necessary.

To further reduce the risk of transient power shorting the board, both regulators in DR2 have enable pins with pull down resistors to ensure they only turn on when they are told to. In addition, there is a switch on DR2 that routes power to the board, acting as on/off switch for the entire system. Lastly, a separate positive regulator for the digital board that is enabled by default to make sure the digital board can always get power as long as the power switch is flipped on.

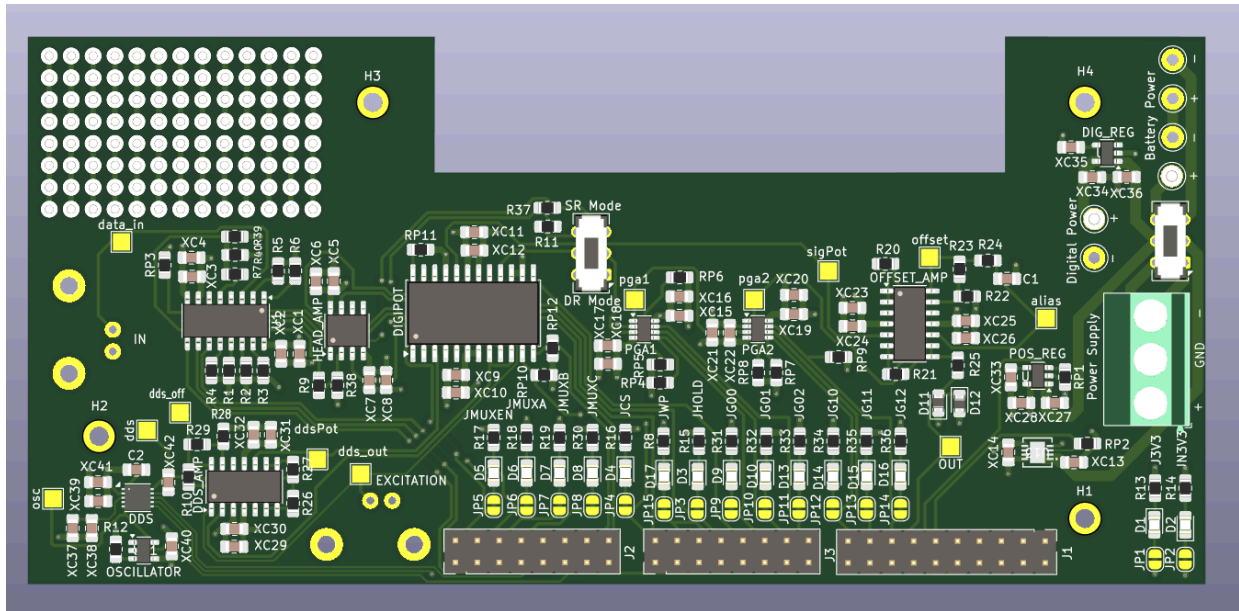


Figure 15: Final Prototype DR2 PCB Layout

3.8 - Excitation Methods

In order to generate a zero offset sine wave, Prototype 6 employed an AC-coupled DAC. However, this system experienced glitches when the processor got too busy. For example 1.5 MHz was the highest rate at which data could be communicated between the DAC and Teensy which was too low to generate a 1KHz high resolution sine wave.

To address the firmware challenges of Prototype 6, the DDS chip was selected as the excitation method instead of the DAC. This choice was made due to its user-friendly interface and reduced need for digital update writes, helping to minimize the communication delays that plagued the previous design (see Section 4.1 Previous Effort of Digitization for more comparisons). When picking the exact DDS to buy, the key parameters were power consumption, low frequency compatibility, and communication protocols. The AD9837 DDS chip was selected because it checked all of the major requirements. It has 3-wire SPI interface, sleep mode and DC operation. In order to generate the MCLK, a 16MHz oscillator chip that runs on 3.3V was added, this is the same one used on the AD9837 development board. The DDS chip can then produce a programmable sine wave with a single digital write, thus it can effectively step down the glitching issue that last year's DAC had.

For DR2, a combination of the DDS chip and a digital potentiometer was used to achieve the same goals while addressing last year's issues. However, the DDS chip can only output sine waves with a positive DC offset, so the output from the DDS must be negated. This is accomplished by feeding the output of the DDS into an offset circuit to provide the negative

voltage needed to center the sine wave at zero volts. Before that though, there is an amplification stage configured as an inverting amplifier, which enables the excitation signal to be amplified from 0.01x to 5x, achieving signals as small as 10 mV or as large as 3.3V.

To ensure proper operation of the DDS chip, a simple power pass-through test was performed on the DDS evaluation board by powering up the board and feeding in waves at SDATA, FSYNC and SCLK pins and reading from the VOUT pin a sine wave. The DDS was programmed to either output a 1000 Hz (AC) sine wave or turn off (DC) using SPI communication. This was tested using an Oscilloscope and a logic analyzer to ensure that the correct bits were being sent over to the AD9837 chip.

3.9 - Layout and Connectors

Two rows of pin headers on the analog board map to the corresponding GPIO and I2C connection section on the Nordic Board. Ribbon cables connect the female pins of both boards to provide stable and easily accessible connections.

The physical design of the system places the analog board on top of the digital board, allowing vertical overlay of connections and easy access for probing during testing. DR2 has four layers: first layer composed of the major signals between the components and to the connector outputs to the digital board, the second layer is ground, the third layer is +3V3 and the fourth layer hosts -3V3 power and non-major signals that route towards the indicator LEDs and the GPIO pins (Figure 15). To ensure stability most of the signal routes used 15 mil and 30 mil for power routes except for the small PGAs and the DDS chip which would violate design rule check. The placement of the components follows the flow of amplification from head-stage amplifier at left edge to offset amplifier at right edge. The capacitors are placed close to their corresponding power pin in order to provide stable power to the components.

Most significant values of capacitors, resistors and component naming was kept on the silkscreen for future reference. The size of the analog board has identical dimensions to the digital Nordic Board in order to save space in the final system container. The placement of the components follows the flow of the signal path. Screw terminals were used for power connection to ease the switch between the batteries and the power supply.

3.10 - Single Rail Design

On the analog side the 2024-25 team looked to investigate the viability of and design a single rail analog board. Section 7.2 details processes from Site Visit 2 to confirm that SR1 is viable for bug feeding and is worth the investigation in the spring and summer of 2025.

To get into the electrical theory of single rail, it is predicted that a single rail design that references the excitation voltage and R_i resistor to 1.65V instead of ground (Figure 16) will be electrically equivalent to the Legacy AC-DC EPG. The insect should feel the same voltage across it as in the Legacy AC-DC EPG, and should therefore exhibit the same feeding behavior. This change would mean the whole signal chain would have a voltage range of 0 to +3.3 V instead of -3.3 to +3.3 V, essentially treating 1.65 V as ground and eliminating the need for a negative voltage supply.

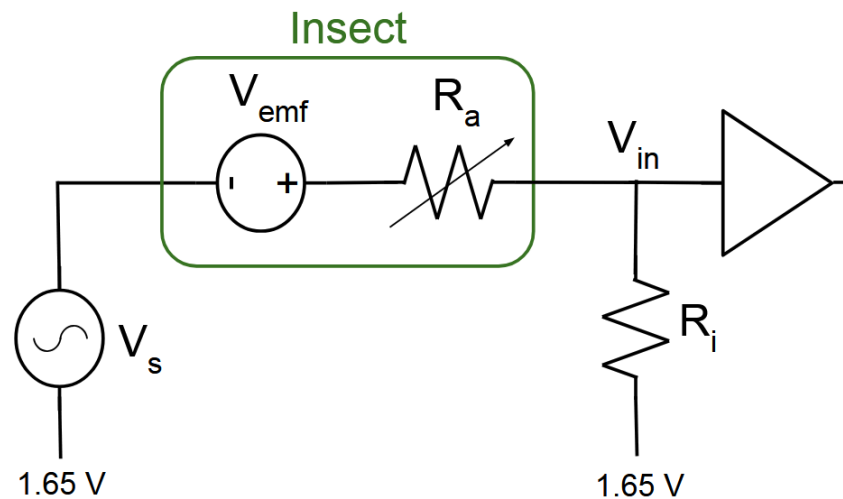


Figure 16. Single rail insect-plant circuit model. The only difference from Figure 2 is replacing the voltages that the excitation source and R_i are referenced to: 1.65 V instead of ground (0 V).

This advantage to this single rail system is that it allows the analog board to be shrunk significantly. Specifically, a single rail board could make use of chips that operate on a single supply, opening the door to a slew of optimized chips designed specifically for this kind of application. For example, to achieve a gain of 21,000 needed to emulate the Legacy AC-DC EPG, the current dual rail design has PGAs which have gains in the thousands and have high resolution. However, they take up lots of valuable board space and need six total GPIO pins to operate. A single rail revision could make use of a differential amplifier called the PGA308. This chip is a programmable analog sensor signal conditioner and includes functionality for amplification, offset selection, and gain select, both coarse and fine. Not only could this chip replace the PGAs, but the offset circuits and potentiometer amplifiers as well.

In addition, a single rail revision would replace the DDS and digital potentiometer combination with just a DAC which can largely reduce the board size. Additionally, higher resolution and granularity would be available to users - up to 16 to 18 bits when the DAC is introduced. Also, the negative regulator would be taken out since there would be no negative power rail, and the

digipot would be replaced with another DAC controller to control the excitation. A JFET buffer could also be added to stabilize the signal before entering the PGA308. The team developed an initial schematic in spring 2025 which will be further developed and shipped in the summer of 2025.

Overall, the single rail design would have a simplified design with reduced components, as seen in Figure 17. This would result in lower power consumption, allowing for the analog board to be powered from the digital board's coin cell, significantly reducing mass and size.

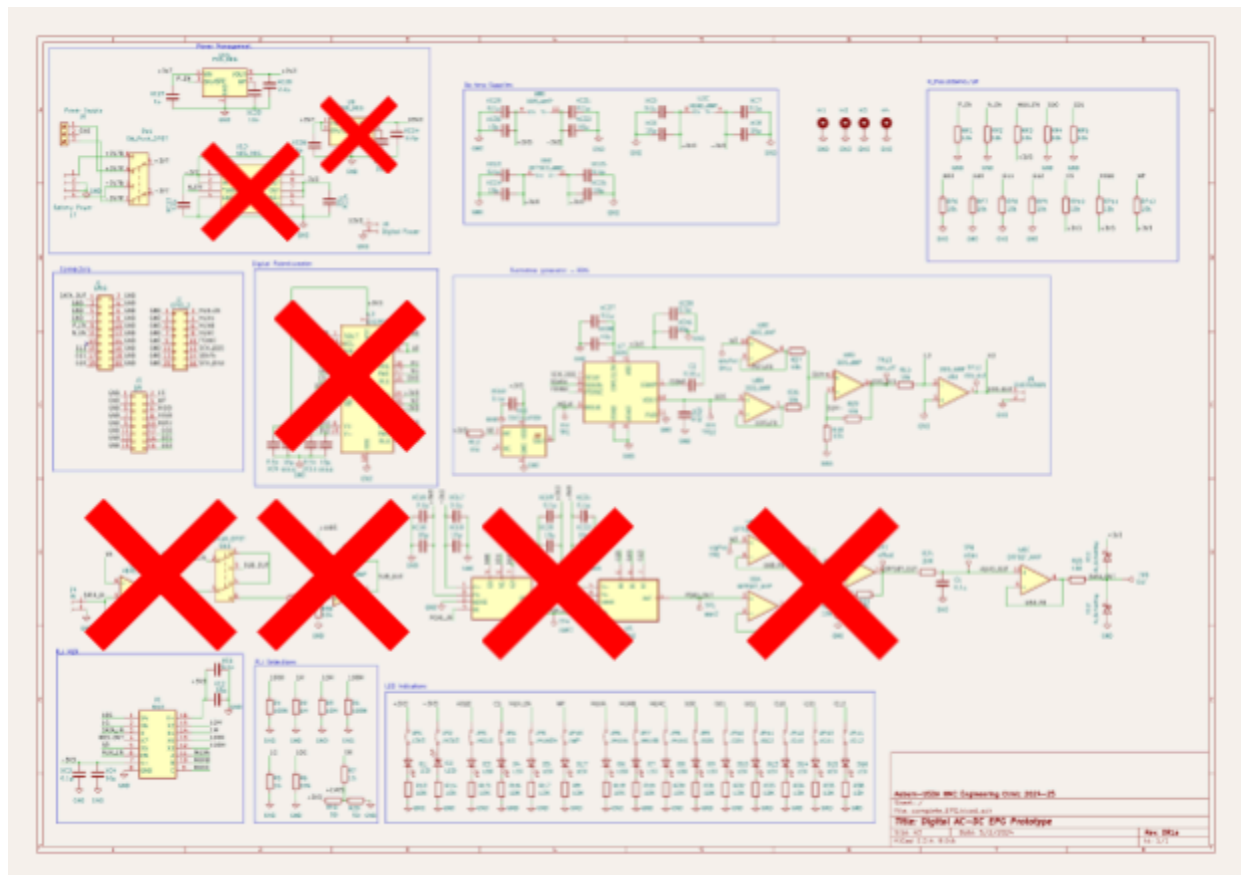


Figure 17: Schematic of DR2 showing which components will no longer be needed in a single rail redesign

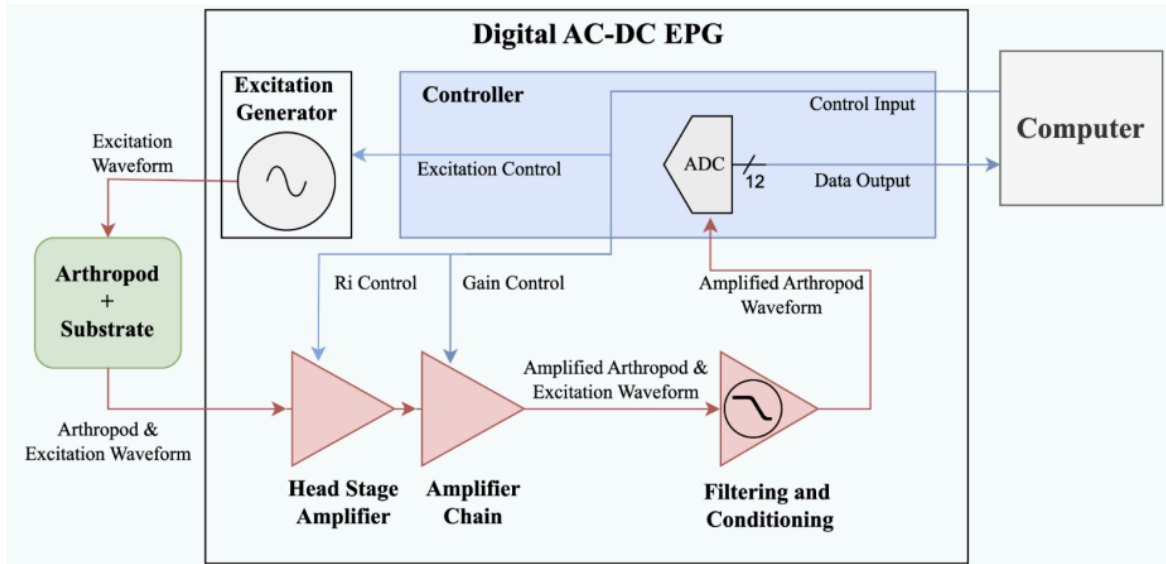


Figure 18: Block diagram of the 2023-24 Auburn-USDA Engineering Clinic Team's whole system [10].

Section 4: Digital Board & Firmware

This section begins with challenges present in the 2023-24 Teensy 4.1 digital board and then delves into how these challenges were addressed in the 2024-25 nRF5340 development kit board including a board trade study, bluetooth communication, ADC and GPIO controls.

4.1 - Previous Efforts of Digitalization

As discussed in Section 2, both Prototype 6 and DR2 had a digital controller. Figure 18 shows a block diagram of the Prototype 6 Digital EPG. In 2023 the digital controller (the controller block in Figure 18) was implemented using a Teensy 4.1 microcontroller. This microcontroller did not support wireless communication, which required DR2 to use a different digital board. The selection of that board is discussed in Section 4.2.

In addition to enabling wireless communication, there were other areas for improvement from last year. The 2023/2024 Auburn-USDA Engineering team report suggested that the digital board and firmware needed to support improved digital amplitude selection, firmware issues, and interference between SD communication and excitation generation. The digital controls didn't check to see if an inputted gain value was in the acceptable range. This was fixed by adding in the digital potentiometer amp which allowed for digital control of the excitation amplitude (see sections 3.2, 3.3, 4.6, and 4.7 for more details). The 2023/2024 Auburn-USDA Engineering Clinic team also ran into problems with only being able to send one datapoint at a time via the DAC versus in batches which slowed down communication (see GPIO

CONTROLS section for improvements). Since SD communication was prioritized last year over the excitation generation and there were delays of 7 ms in the SD communication, that lag delayed excitation generation signaling from the DAC.

4.2 - Board Trade Study

In order to address last year's concerns regarding the lack of wireless communication, a trade study on boards was completed that contained Bluetooth Low Energy (BLE) compatibility, SD communication, an antenna port, ADC, power consumption, and software support and documentation. These features were selected to meet the requirements in Section 2 under Bluetooth architecture. The ESP32 or nRF5 series are the only serious Bluetooth chipsets with readily available development boards. They were investigated to see how they incorporate wireless communication while also having an adequate ADC to sample from the analog chain [10]. After comparing the pros and cons of both, the Nordic nRF5340 was chosen because it has many GPIO pins (48 overall GPIO pins and 23 directly accessible), I2C interfaces (P1.02 and P1.03) available, and it is better designed for low power consumption (Table 2). The Nordic board has four different power supply options: USB (5V), LiPo battery (2.5-5V), external supply (1.7-3.6V), and coin cell battery (3V). Another feature included is an onboard ADC with up to 8 channels, 12 bits, and up to 200 kbps sampling. This board does not have a SD card reader, but it does have non-volatile storage (NVS) for storing and reading data in flash memory. Lastly, Nordic has its own software development kit which includes sample code written in C and documentation.

BLE	Pro	Con
Espressif (ESP32 Series)	- Cheaper - More popular → larger community and resources - Environment with examples of use of the communication device - Various communication peripherals	- Problematic consumption (even with low power mode), severe on battery system - Reports of long-term issue of halting after stack overflow - Current peaks > 250mA during radio transmissions - Documentation lack/not in English
Nordic (nRF5x series)	- Many iterations, focus especially on low-power → suitable for battery supplied application - Easy BLE interface and pin assignment - SoftDevice: radio functions & Segger Embedded Studio: ease debugging	- More expensive - Usually separates WiFi and BLE

Table 2: Trade Study on Espressif and Nordic development boards with BLE compatibility.

Custom Service UUID	445817D2-9E86-1078-1F76-703DC002EF41
Read Characteristic UUID	445817D2-9E86-1078-1F76-703DC002EF42
Write Characteristic UUID	445817D2-9E86-1078-1F76-703DC002EF43
Device Address	D5:CA:E6:A3:9A:09

Table 3: Hardcoded values of the UUIDs and device address

4.3 - Zephyr Operating System

The integrated development environment (IDE) used for programming the board was nRF Connect for Visual Studio Code (VS Code) [15]. This IDE enables code writing, project building, and program flashing onto the digital board. The nRF Connect SDK (Software Development Kit) [14] is the primary code framework, offering an extensive set of libraries, drivers, and sample applications specifically designed for Nordic Semiconductor's boards.

The nRF Connect SDK integrates the Zephyr Real-Time Operating System (RTOS) [15], a lightweight and scalable RTOS ideal for resource-constrained devices similar to what is needed for this project. Zephyr provides important OS features such as task scheduling, memory management, and inter-process communication. These features help write modular, efficient code that meet the requirements of the project.

Using this SDK, the board can be flashed with code from a compatible computer via USB, utilizing the on-board SEGGER J-Link debugger. This setup enables a quick and straightforward development process that follows a simple workflow: write the code using the SDK, compile it, and then flash the successfully compiled code to the board.

4.4 - ADC Control

A key function of the digital board is to capture the analog waveform produced by the analog board. This is achieved by the on-board Analog-to-Digital Converter (ADC).

The analog waveform from the analog board is sent to the digital board where it is digitized using one of the digital board's pins that supports the on-board ADC. The ADC on the digital board is capable of sampling at rates up to 200,000 samples per second with a 12-bit resolution [4]. Once the waveform has been digitized, the data is able to be plotted for visualization as well as stored for future reference.

The ADC's sample rate can be configured arbitrarily, but for testing purposes, it was set to one sample per second using a 1-second delay between sample requests.

4.5 - Power-Up Sequence

The digital and analog components of the system are both powered by LiPo batteries that connect to the analog board and controlled by supply switches to turn the system on and off. This separation introduces the potential for destructive sequencing at startup, where one side may power up before the other, potentially damaging components or causing unexpected behavior as described in Section 3.7.

To address this, the analog board in this year's design contains enable pins for each regulator controlled by the digital board, which manages the power-up order. This ensures that no analog signals are sent to the digital side until the digital board supplies an enable signal to the analog board. This safeguard prevents premature communication between the two sides, protecting both analog and digital components.

An additional measure taken to ensure a safe startup sequence was to configure each GPIO pin on the digital board with a pulldown resistor (the default configuration for all pins). This ensures that all GPIO pins are low before the analog board is powered up. By keeping the GPIO pins low, it prevents any power from being sent to the analog board while it is still powered off. This protects the analog components from potential damage caused by power from the digital side.

The power up sequence will follow an order that resembles the following:

1. Power on the digital side of the system.
2. Ensure all GPIO pins are low (set to 0/ground)
3. Enable the power sequencer to initiate the power-up of the analog board.
4. Wait some time for the analog side to stabilize/power-up.
5. Initialize the chips on both the digital and analog boards using SPI and GPIO writes to configure them to some default operation.

4.6 - GPIO Control for PGAs

To maintain high amplification in gain control, programmable gain amplifiers (PGAs) were included in the design of DR1, continuing the approach used by the 2023-24 engineering team. Each PGA can be digitally controlled via 3 GPIO pins per chip, allowing for precise adjustment of the gain. Since the analog board uses 2 PGAs, a total of 6 GPIO pins are dedicated to controlling the amplifiers from the digital board.

These GPIO pins are configured to pull down by default, meaning they initialize to ground at startup. This configuration ensures that the system behaves predictably at power-up, particularly because the order in which the components power on is not guaranteed. By pulling the GPIO pins to ground initially, sending power to components that may not yet be powered on is prevented.

The pull-down behavior of these GPIO pins can be modified through the SDK, providing flexibility in how the system initializes. However, for safety and reliability, the default pull-down ensures that no unwanted signals are sent to the analog board during the startup power sequence.

4.6.1 - How To Control PGAs

Each PGA is controlled by a combination of three GPIO pins, and the state of these pins determines the level of gain amplification. Figure 19 below illustrates how the states of GPIO pins (G2-G0) influence the nominal voltage gain of one of the PGAs, the LTC6910-3 [1].

Table 3. LTC6910-3

G2	G1	G0	NOMINAL VOLTAGE GAIN		NOMINAL LINEAR INPUT RANGE (V _{p-p})			NOMINAL INPUT IMPEDANCE (k Ω)
			Volts/Volt	(dB)	Dual 5V Supply	Single 5V Supply	Single 3V Supply	
0	0	0	0	-120	10	5	3	(Open)
0	0	1	-1	0	10	5	3	10
0	1	0	-2	6	5	2.5	1.5	5
0	1	1	-3	9.5	3.33	1.67	1	3.3
1	0	0	-4	12	2.5	1.25	0.75	2.5
1	0	1	-5	14	2	1	0.6	2
1	1	0	-6	15.6	1.67	0.83	0.5	1.7
1	1	1	-7	16.9	1.43	0.71	0.43	1.4

Figure 19: LTC6910-3 PGA control table

4.7 - SPI Communication for Digital-Potentiometer & DDS Chip

To address last year's issues with the DAC and gain resolution, a new digital potentiometer and DDS chip were added to the design. These components require SPI communication to dynamically configure and control their operation.

4.7.1 - Digital Potentiometer

The digital potentiometer is controlled via SPI using instructions outlined in the datasheet [2]. SPI1 on the digital board was enabled and configured for communication with the digital potentiometer. Communication with the chip works by sending an instruction byte in the SPI protocol, which selects one of the four potentiometers on the chip and writes to its corresponding wiper register (WRC). The primary focus is on modifying the WRC to adjust the potentiometer's current step and set the desired mode of operation. This is a 3-byte instruction which is described in more detail in the datasheet [2].

4.7.2 - DDS Chip

The DDS chip uses SPI communication to configure its settings. Similar to the digital-potentiometer code, various instructions can be written to the DDS chip via SPI to configure its operation and dynamically control its settings as needed.

4.8 - BLE Communication

Wireless communication between the Clinic-v2 firmware and the Clinic-v2 frontend via BLE was integrated. BLE was chosen because it is power efficient and reduces noise from floating point measurements. This year's team focused on the host interface which includes the Generic Access Profile (GAP). GAP is responsible for setting up the connection between other devices

and advertising data in either a broadcast or a connection-oriented approach. In this case, a connection-oriented approach is used. In this approach one connection is made between a server and a client BLE device as opposed to a server being able to communicate with multiple devices [12]. This interface also includes the Generic Attribute Profile (GATT), which is a collection of services and characteristics that include the data being read or written by other devices. Each service and characteristic contains a unique 128 bit Universally Unique Identifier (UUID). To learn more about the system please check the tutorial listed in [18].

When implementing BLE, the library `bluetooth.h` is used. Certain connection parameters can also be modified within the `conn.h` file if users wish to increase the data throughput of the BLE. These parameters include the connection interval, the number of packets per interval, and the packet length. The connection interval dictates how often the server and client communicate with each other and typically has a minimum of 7.5 ms and increases in 1.5 ms increments. The packet length is typically 20 bytes. The throughput is described in Equation 1 [5, 6, 10].

These libraries are part of the Zephyr operating system which is discussed in Section 4. A custom service with two custom characteristic values using an online UUID generator was declared. One of them is a notify characteristic and the other is a write characteristic. With a custom service external devices, such as a computer running a Python script containing the frontend, can scan for and subscribe to the custom service. Then the computer can be notified of incoming data, such as voltage values. By having a write characteristic, data can also be read from Python and written to that characteristic so that it is stored on the host side. The device address, service, and characteristic UUIDs are hardcoded which ensures that the right device is connected on the frontend side (Table 3).

In terms of server/client communication, BLE is initialized first as a server and then data from the custom service gets advertised to potential clients which are nearby devices that can scan for the advertised services. In this year's case the client is a local computer running the Clinic-v2 frontend. In the `bluetooth.c` code for the BLE server, an advertisement structure sets up the service connection and notify/write permissions for the characteristics. BLE is initialized in the `main.c` script, prompting the advertisement to begin. It calls functions that read in data from the ADC and from the UI. In the ADC thread running in `main.c`, where the voltage and timestamp data is stored in a buffer, a function from the Bluetooth code is called so that the data can be transmitted and stored to a separate buffer on the BLE side. Every time that function is called the frontend (client) becomes notified of incoming data and receives it. Then the data is able to be plotted. The Clinic-v2 frontend is responsible for creating the UI and communicating with the BLE server. To scan for the service and read in the characteristic data the `BleakClient` library was used along with the Python library `asyncio` to implement threading. The script has a function that can write data to the server's BLE custom service (see SOFTWARE AND UI for

more details). On the BLE side that written data is processed and used to control chips on the nrf5340 dk.

$$\text{Throughput} = \frac{1000 \text{ ms} * \text{number of packets per interval} * \text{packet length}}{\text{Connection Interval (ms)}}$$

Equation 1: The amount of data transmitted through BLE depends on the number of packets per interval, packet length, and connection interval [12].

Section 5: Frontend / User Interface

The frontend/UI responsible for visualizing the collected data and controlling an EPG system was inherited from last year's computer science clinic team. This existing software and codebase served as the foundation for this year's project [11]. The UI was developed in Python using PyQt5, a Python library for creating graphical user interfaces (GUIs) based on the Qt toolkit [10]. The UI responsible for visualizing the collected data, and controlling an EPG system was inherited from last year's computer science clinic team. This existing software and codebase served as the foundation for this year's project which can be seen in Figure 20 [11].

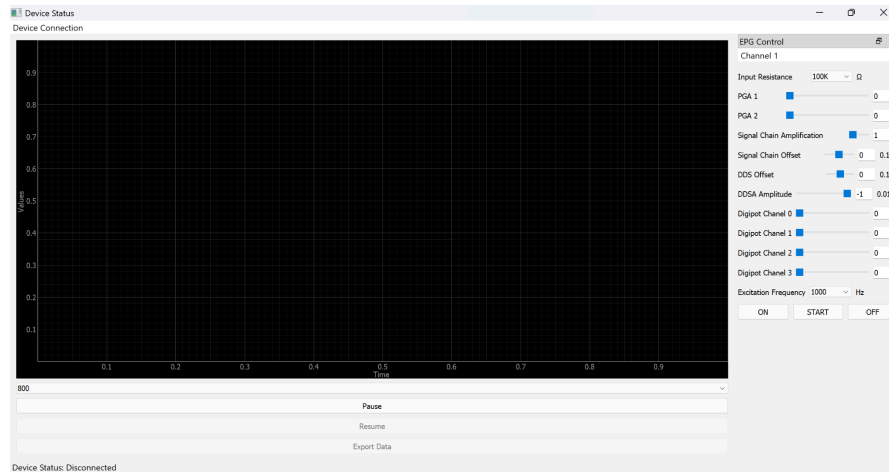


Figure 20: The Clinic-v2 Frontend with the controls on the right and the plotting software on the left.

5.1 - Connect status - Bluetooth integration

In the Clinic-v2 frontend, one of the functions that is called when the plotting window is initialized is `_createConnectivityStatusBar()`. This function is responsible for creating a `QStatusBar` and a thread for running the BLE connection process. This thread also checks the connectivity status of the device which is displayed in the status bar. With this status bar users

are able to see whether or not the device is connected to bluetooth. Users can choose to connect to bluetooth using the “Device Connection” menu. Selecting “Connect Bluetooth” starts the thread and allows BleakClient to start scanning for devices to connect to, which in this case is the nrf5340 dk. Once Bluetooth is connected users can control the system using the EPG Control window located on the right side of the window.

5.2 - Available sliders

The EPG Control window contains ten sliders, seven buttons, and two drop down menus for users to interact with when controlling the chips on the nRF5340-dk. The ten sliders are used to control both PGAs, the signal chain amplification and offset, the DDS offset and amplitude, and all four digipots. Users can select the input resistance and DDS frequency values from the two drop down menus titled “Input Resistance” and “Excitation Frequency” respectively. Lastly the system can be turned on or off using the “ON” or “OFF” buttons respectively. Data starts streaming to the UI once the “START” button is pressed. Every time a value is changed on a slider, button, or drop down menu, a write function is called that is responsible for writing the control bits to Bluetooth. Different messages are sent over depending on which control is activated. These messages are then parsed by Bluetooth on the nrf5340 dk (see Section 4.8 for more details).

5.3 - Waveform viewing

Waveform data is plotted using the PyQt5 library, which provides a range of built-in features like mouse-controlled graph manipulation that significantly simplified the plotting process compared to last year’s graphing process. To ensure smooth performance, only the data currently stored in a circular buffer is displayed. Users can configure the number of points stored in this buffer; however, increasing the number of points may lead to performance degradation. The team found that plotting up to 1,000 data points (equivalent to 10 seconds of data at a 100 Hz sampling rate) maintained a responsive user experience.

An enhancement for future development is implementing dynamic downsampling based on the current zoom level. For example, when viewing 60 seconds of data, displaying all 100 points per second (at 100 Hz) is unnecessary, as individual data points would be indistinguishable. In such cases, selectively reducing the number of visible points could optimize performance without sacrificing visual clarity. Conversely, at higher zoom levels (e.g., a 1-second view), showing all data points is necessary to preserve waveform detail. Implementing this approach would also enable the display of long-duration waveform data (like an hour of continuous recording) without the application crashing.

The Computer Science team explored this feature and developed a functional prototype that included a full-history waveform display with dynamic downsampling, and demonstrated functionality using a simulated 100 Hz sine wave generator. Although functional, due to time constraints leading up to the final site visit, this feature was not fully integrated into the final version of the graphing application. However, the underlying code is in place, and integration into the main application should be a manageable task for future teams.

5.4 - Data export

Data from the EPG is transmitted in (timestamp, voltage [mV]) format and can be saved into a CSV file while the recording is paused. Pausing the recording will not stop data collection, it just stops the graph from plotting.

Section 6: Mechanical Setup

The analog PCB, digital board, and batteries needed to be assembled into one easy to use unit that would be sturdy, shield against noise, and portable. In order to best achieve these goals the team decided to customize the product housing.

Since the analog and digital board needed to be grounded into the chassis, the housing needed to be made out of metal. In order to accommodate necessary ports for wired operation, bluetooth antenna, and BNC ports, the mechanical housing needed to be custom machined to the specifications shown below in Figure 21. Given the imperative for shielding from noise, the team manufactured both a steel and aluminum chassis for the first site visit. Steel is able to shield from near and far field magnetic fields while aluminum is only able to shield far field magnetic interferences.

During the first site visit both the steel and aluminum housings performed comparably. As a result, the team was able to consider them both without compromising on the shielding ability of the housing.

Upon analyzing the costs and benefits of both options, the team concluded that steel is heavier, and is not available off-the shelf which meant that the entire housing would have to be custom built out of sheet metal while the aluminum housing would be lighter and could be purchased off the shelf. Given these considerations, the team decided to go with an off the shelf aluminum housing from poly case [22].

The aluminum casing purchased was machined to include two slots 1.25 inches in height and 0.5 inches in diameter for the excitation and host voltage BNCs, a port 0.5 inches tall and 0.6 inches in width for wired connection, and a 0.4 inch hole on the lid for the bluetooth antenna.

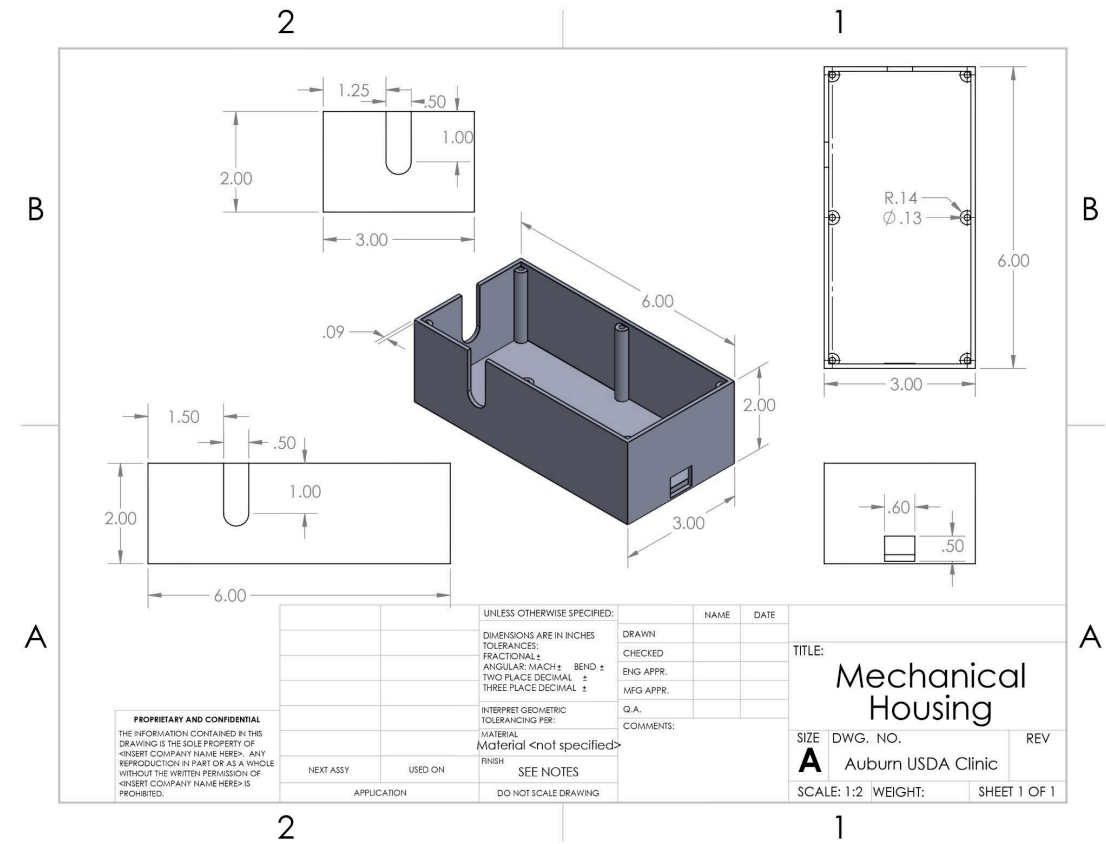


Figure 21: Solidworks drawing of mechanical housing

The final mechanical housing used for the second site visit can be seen below in Figure 22. Shorter ribbon cables and flatter BNCs were selected in order to minimize the size of the prototype. A bolt was attached to the bottom and wrapped in an insulating material so that the housing could be attached to the ring stand mount without conducting the noise attracted by the large metal apparatus. The bolt could be removed so that the housing could be rested on any lab surface or strapped to an animal such as a cow in the future. The custom housing developed by the team ensured minimal dead space in the chassis helping to cut down on size, was lightweight, easy to use and effective at shielding the system from noise.

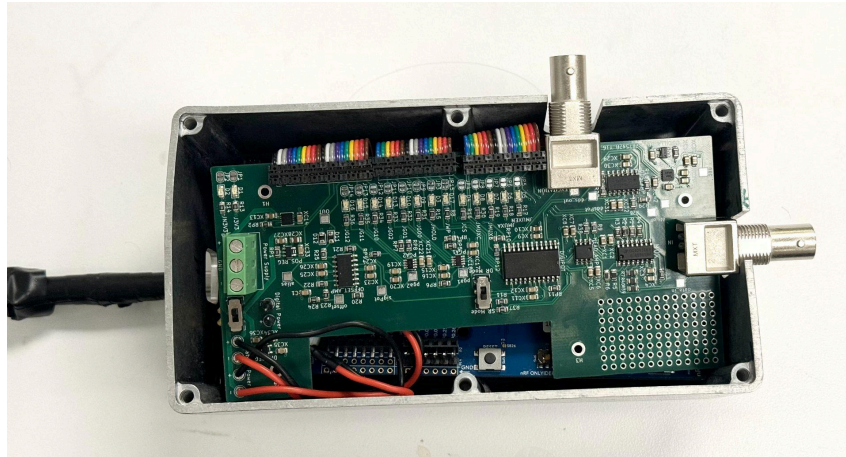


Figure 22: Mechanical housing with prototype inside and BNC connectors extruding from box for probe connections

Section 7: Site Visit Results

The 2024-25 team took two site visits to Parlier, California to the USDA Agricultural Research Department to test various prototypes and theories. This section will highlight site visit experiments and results.

Site Visit 1 was taken in the first week of March and the main goal was to test DR1a using UART connection on aphids and investigate optimal setups for noise limitation. Site Visit 2 was taken in the last week of April and the main goal was to test DR2 with BLE connection and confirm the viability of single rail for bug feeding.

7.1: Site Visit 1

Site Visit 1 included various experimentation using analog board DR1a with UART connection. The team used a barebones version of Frontend-v2 for basic waveform viewing and data saving. During the first site visit, the team investigated various noise setups to see how they affected noise for waveform viewing. The team completed a qualitative analysis with the liaisons present, who were able to provide immediate feedback about which setups worked well compared to others.

The team investigated the following comparisons for noise tests: (1) Inside faraday cage versus outside faraday cage, (2) steel versus aluminum mechanical box, and (3) ring stand versus no ring stand. Surprisingly, the prototype worked better outside the faraday cage, with significantly less noise. However, the faraday cage not being necessary is useful for entomologists as it opens up the possibility of many different and new experimental setups.

For the mechanical boxes, the liaisons noticed marginal improvements with the steel box, however it did not perform that much better than the aluminum box for this to be significant.

After the faraday cage test, the team experimented outside of the cage with the ring stand and without, and found a significant improvement in noise when the prototype was separated from the ring stand. The team hypothesized that the ring stand acted as a long antenna and drew extra noise.

Once noise tests were conducted, the site visit culminated in collecting aphid feeding waveforms. These results were collected using the setup with the best signal to noise ratio, which was the aluminum box outside the faraday cage without the ring stand as shown in Figure 23.



Figure 23: Experimental setup for Site Visit 1 collection of aphid waveforms

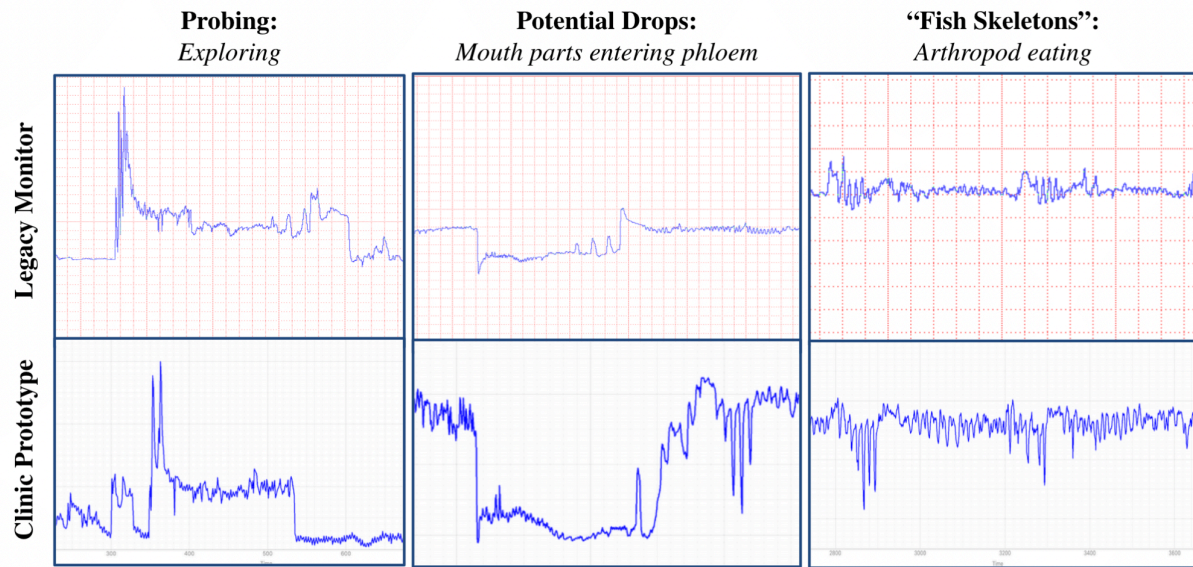


Figure 24: Comparison of aphid waveforms from Clinic Prototype DR1a and UART connection versus the legacy monitor.

The 2024-25 team collected data and was able to pull out various characteristic waveforms as shown in Figure 24. The direct comparison between clinic prototypes to aphid feeding patterns is crucial as aphid feeding is the gold standard for EPG technology. As shown in Figure 24, the prototype demonstrated close performance to the legacy EPG for probing/exploring, phloem feeding, and eating behaviors.

7.2: Site Visit 2

Site Visit 2 included experimentation with DR2 to also test bluetooth capabilities for controlling analog settings and viewing data completely wirelessly. The team also investigated a single rail setup to confirm the biological viability of differential measurement and its effect (if any) on aphid feeding.

Bluetooth functionality was maintained for a maximum of 46 feet of distance from the antennae and through three walls at the USDA Agricultural Center. The team was also able to adjust controls on the analog board and view data in real time completely over bluetooth.

For feeding waveforms, the team looked to both replicate aphid waveforms as seen in Site Visit 1 and also test functionality with sharpshooters, which feed using an AC excitation voltage. Figures 25, 26, and 27 show aphid feeding waveforms similar to the ones shown at Site Visit 1. All three characteristic behaviors – probing, potential drops, and eating – are directly comparable to DR1a and legacy monitor results, suggesting the success of DR2 and bluetooth combination.

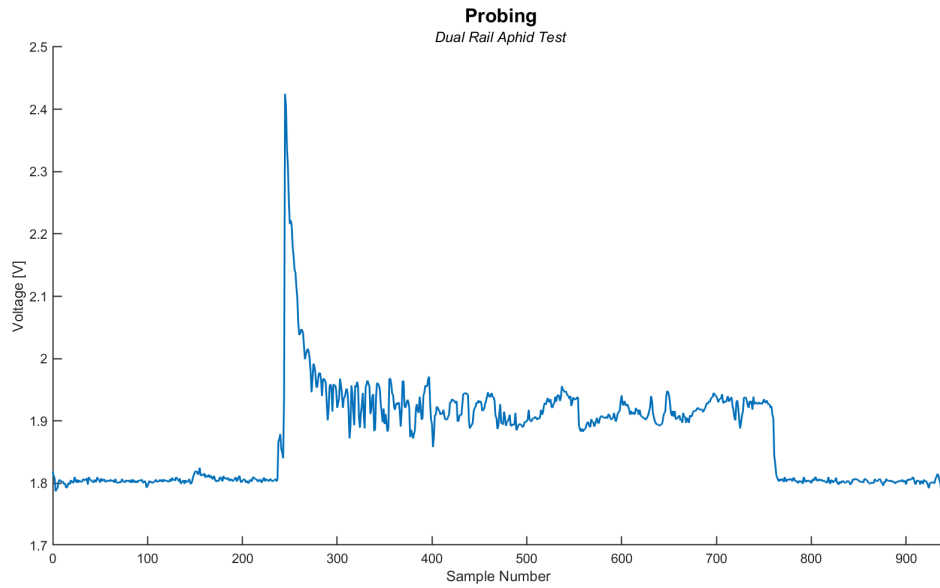


Figure 25: Probing waveform produced by DR2/Bluetooth prototype

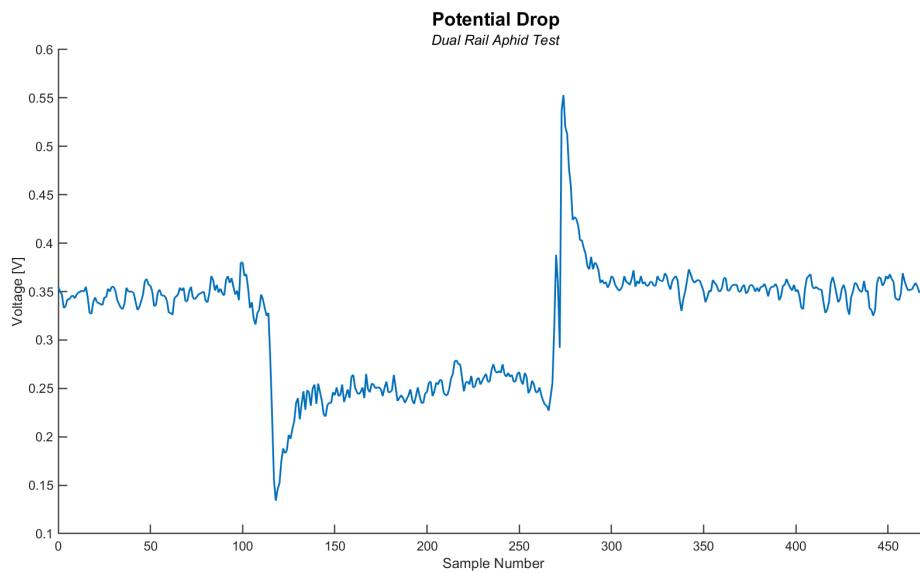


Figure 26: Potential Drop produced by DR2/Bluetooth prototype

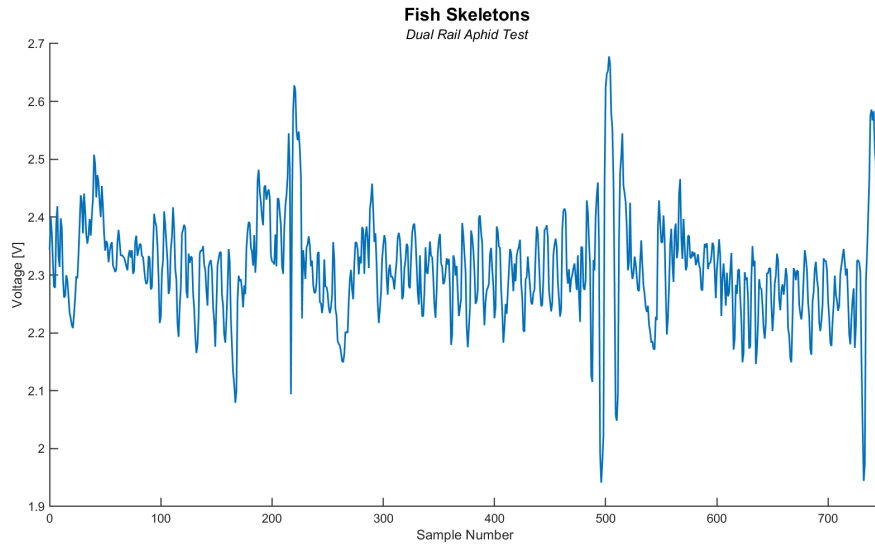


Figure 27: Fish Skeletons produced by DR2/Bluetooth prototype

In addition to aphid waveforms, DR2 looked to collect proof of sharpshooter feeding by using an AC excitation signal. Figure 28 shows comparable X waves for the dual rail sharpshooter test, as confirmed by liaisons over the site visit.

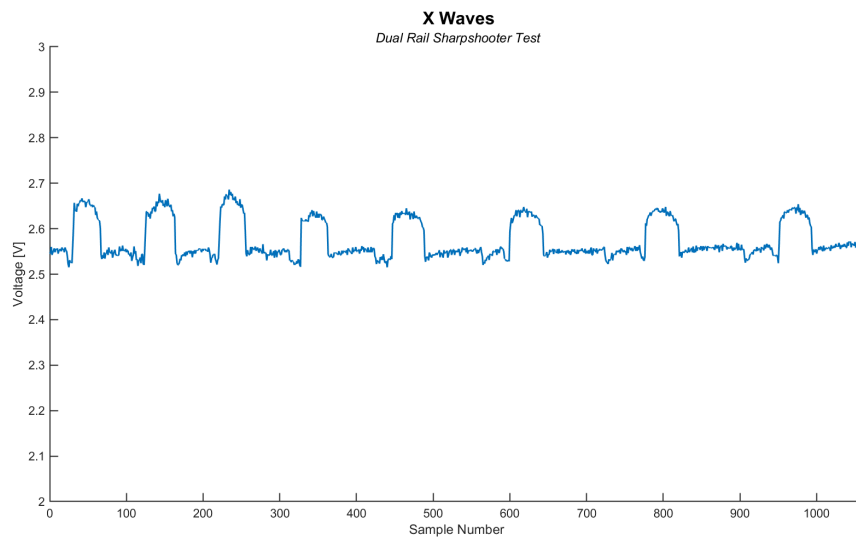


Figure 28: Sharpshooter X waves produced by DR2/Bluetooth prototype

Another important goal of Site Visit 2 was to investigate the possibility of a single rail design. As shown in Sections 2.2 and 3.10, a single rail design would significantly decrease the power consumption and area on a PCB. However, the use of differential measurements raises new

biological questions, and thus it needed to be confirmed whether aphids would feed under such conditions at Site Visit 2. A simple experimental setup was soldered onto the prototyping area of DR2 in order to test the differential measurement. A simple subtracting circuit was placed after the head stage and before the PGA amplification in order to remove the 1.65 V offset before it was amplified downstream. This was done so that only the signal of interest would be amplified in the following gain stages as opposed to the 1.65 V. The results from the single rail test are shown in Figures 29 and 30 for potential drops and Fish Skeletons, thus confirming that aphids are not negatively affected by the differential measurement and a single rail design is a viable option for future work.

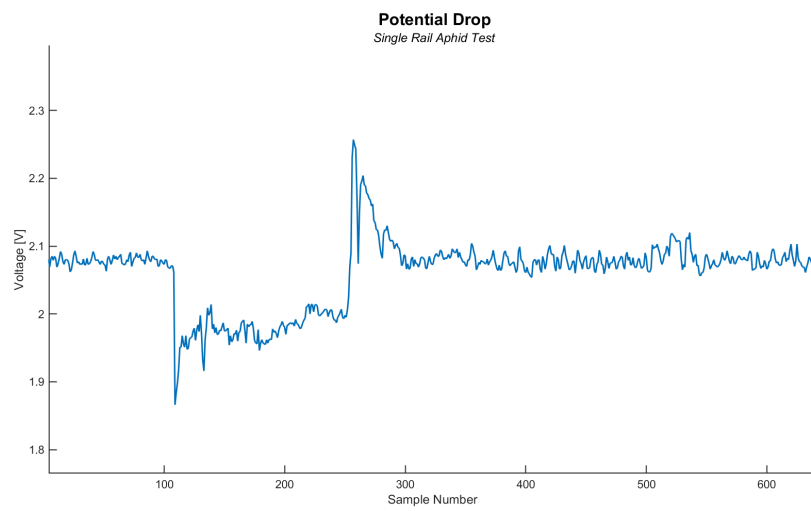


Figure 29: Results from single rail testing, showing a potential drop

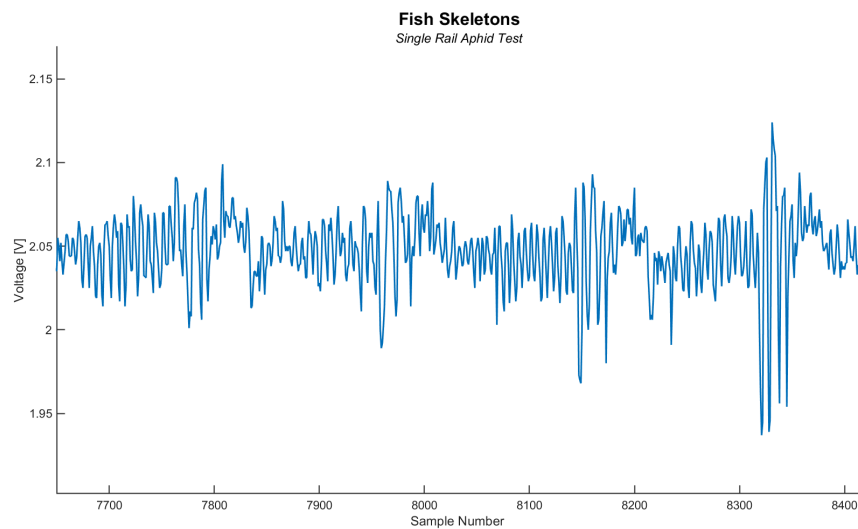


Figure 30: Results from single rail testing, showing fish skeletons

Lastly, further testing was completed to continue the noise investigation from Site Visit 1. The first was discovering sources of noise in an experimental setup, which involved introducing external electronics such as chargers to replicate an entomologist's setup, which often includes cameras, microscopes, and other cabling. The team discovered during their experimentation that two prong plug-ins introduce significantly more noise than three prong plug-ins. Additionally, the prototype functions as normal with relatively noiseless results as long as all cabling or wires are 6 inches away from the gold wire/bug attachment that protrudes from one BNC.

Section 8: Project Accomplishments and Future Work

This section will detail accomplishments and preparation for Summer 2025 research.

8.1 - Accomplishments

A main deliverable for the fall semester was to pick up where the 2023-24 clinic teams had left off to address issues with their final prototype. The adjustments that were involved with that include developing an updated analog board, revising the digital board choice to implement more robust ADC, analog control, and bluetooth capabilities, and improving the software to make it easier for entomologists to interface with the digital and analog boards.

DR1 was successfully shipped in mid-November, and included an excitation generator, a head stage amplifier, and gain and offset amplifier chains. The nRF5340-DK digital board was chosen and a 12-bit resolution ADC was implemented, along with starting bluetooth and non-volatile storage. Additionally, functionality to interface with the PGAs, digital potentiometer, and DDS was implemented. The bluetooth module can also connect to and transmit data to the user interface. On the UI side, bugs were resolved from the 2023-24 CS team's project, including bluetooth device connectivity status and easy-to-use analog adjustments.

The spring semester focused on finalizing the analog design and integrating it with the work done on the digital board and a mechanical housing in time for the 2 site visits in Parlier, CA. This also included Bluetooth integration and a complete UI that can work free of bugs and other interferences in a robust prototype. Lastly, the team evaluated the possibility of a single rail design to prepare the summer team for further research.

The clinic project ended with a successful prototype DR2 in combination with Bluetooth capabilities and a finalized UI for live data plotting, controlling, and data exporting functionalities. Site Visit 1 and 2 showed acceptable waveforms for both aphids and

sharpshooters, with data being collected completely wirelessly and even in other rooms. The final prototypes worked well outside of the faraday cage and without the ring stand, widening the scope for experimental possibilities.

8.2 - Future Work

On the analog side, the most important next steps for the Summer 2025 team will be to design, develop, and test a single rail board, especially given that it proved to be biologically viable at Site Visit 2. Additionally, formal noise tests using traditional FFTs and Waterfall FFTs will help the team with further analysis to investigate sources and solutions for noise rather than merely a qualitative analysis with the help of the liaisons. Formal battery life tests will also allow the team to decide clearly for tradeoffs between battery size and capacity. On the power management side, it would also be useful to have more LEDs to denote battery charging. In addition, the jumper wiring setup needs to be rethought in order to make the integration with the digital board more efficient and cleaner, including JST connectors for the batteries. Lastly, future work could also implement chopping which describes the switching between amplifying the bugs resistance and the bugs voltage representation.

The Clinic-v2 CS team suggests that the visualization and control software developed on the engineering team should remain a backend tool solely for the engineers to be able to validate the hardware in a barebones application. Work still remains for creating a front facing UI such as debugging and optimizing the Clinic-v1 frontend, or some other front facing UI that is more user focused and easy to use by entomologists, including things like annotation features and sliders with less granularity (since they shouldn't need to control individual chips like the engineering team).

Engineering Team UI – For Internal Team Use

However, there are still some bugs present in the current system that should be addressed in order for the system to be easier for engineers to work with. Right now the sliders on the EPG controller can only handle integer values when a floating point range of -3.3 to 3.3 is required for the DDS offset, DDS amplification, and Single Chain amplification sliders. Currently the output values of these sliders are multiplied by 0.1 so that the correct values are sent back to the digital board, but this can be confusing for users who see the number 33 on the slider instead of 3.3. In the future a floating point value slider should be implemented instead of an integer slider.

Another change to be made on the UI is integrating the UART communication protocol so that users have the option of selecting which communication protocol they want to use (UART or Bluetooth).

When starting data transmission, the current “START” button enables data to be read from the digital board upon initialization of the board. However, if the board is not reset and Bluetooth has been running for a while when the user presses the “ON” button the data starts plotting without the “START” button being pressed. This can be fixed by either implementing the equivalent of a reset command or halting the ADC thread after the “OFF” button is pressed so that the next time the system is powered up data transmission will not start immediately.

Mechanical Housing should be further customized in the future to improve prototype performance. While the housing shields the system well from noise, the excitation stubs that wire to the arthropod are exposed to any magnetic interference and contribute most to the little noise we see in the data. In order to mitigate this a mini faraday coil could be implemented around the stub.

The housing should also be tailored to the different use cases, for example a lighter and more flexible housing could be useful for upcoming calf trials.

Finally, an external on off switch for both power management in the analog board and for the digital board could allow the system to be turned on and off without needing to open the box and disassemble the setup.

Front Facing UI – Designed for Users of EPG

An issue encountered with integration of this year's hardware with the Clinic-v1 frontend was that it had trouble visualizing the thousands of data points collected from the EPG system. Given the engineering team's short staffing of CS majors, they did not have time to debug the Clinic-v1 frontend and optimize it to plot the high amount of data, so work remains on flushing out a complete frontend focused on external EPG user experience similar to the WinDaq software entomologists are familiar with.

Section 9: Conclusion

This document detailed the progress completed by the 2024-25 Auburn-USDA Engineering clinic team, addressing fixes from the 2023-24 analog prototypes, updating the 2023-24 digital board, and improving the 2023-24 software. Analog prototypes were successfully shipped and tested, as well as digital functionality improvements implemented on a newly chosen digital board, together with new software capabilities that together, will improve the data quality and collection of the modernized EPG.

The work completed this year will set the summer team well to focus heavily on iterating over the technical progress with a focus on field testing and single rail.

The Harvey Mudd 2024-25 Auburn-USDA Engineering Clinic thanks the liaisons for their support this year.

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